

Model-level

H1: Composite orientation

$$\text{PVC}_m \rightarrow \Delta_m$$

Aggregate PVC index predicts
cross-model burden effects

$N = 15$ model endpoints
 $\beta_1 = -2.14$, 95% CI $[-6.23, 1.95]$
 $R^2 = 0.09$

Null: no detectable association

Narrative-level

H2: Narrative frame association

$$\text{Frame}_{mi} \rightarrow \Delta_{mi}$$

Interpretive frames correlate with
predicted burden effects

$N = 2,019$ rationale-effect observations
Access barriers $\uparrow$ ($b = 2.36$)
Co. responsibility n.s. ($b = 0.69$)
Coercive backlash $\uparrow$ ($b = 7.26$)

Partial support: 2 of 3 frames

Within-model

H3: Frame manipulation

$$\Delta_{mcir} \approx \text{Frame}_c$$

Assigned governance frame shifts
predicted effect within the same model

$N = 4$ models $\times$ 27 profiles $\times$ 4 frames
Autonomy $\uparrow$ (+4.96), equity $\uparrow$ (+4.84)
Collective obligation $\downarrow$ (−10.88)
adj. $R^2 = 0.77$

Supported (pooled; model heterogeneity)

Does normative-frame sensitivity affect LLM-assisted policy simulation?

Three independent diagnostic tests — each informative regardless of the others.

PVC null (H1) $\rightarrow$ frames partially matter (H2) $\rightarrow$ causal manipulation confirms (H3)